

KWASI GYIMAH-ASANTE

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EDUCATION

Fordham University

August 2024 – May 2028

B.Sc. in Mathematics & Computer Science

New York, NY

- **Relevant Coursework:** Computer Algorithms, Data Structures, Discrete Mathematics, Calculus I, II, III
- **Social Chair, Society for Industrial and Applied Mathematics (SIAM):** Organized chapter events, collaborated with SIAM chapters across other universities, and managed the chapter website

EXPERIENCE

Fordham University – Department of Computer Science

May 2026 – Present

Undergraduate Research Assistant

New York, NY

- Developing a **value iteration** solver for an MDP formulation of quantum network scheduling in **Python**, establishing an exact optimal policy benchmark to systematically evaluate and compare RL algorithm performance under Prof. Xiaolan Zhang
- Modeling quantum network topology as a discrete-state MDP, designing the state-action space, reward structure, and transition dynamics to enable rigorous benchmarking of **reinforcement learning** scheduling policies

PROJECTS

RAG Pipeline | *Python, FastAPI, OpenAI, Anthropic, ChromaDB, BM25, LangChain, Docker*

[GitHub](#)

- Built a **RAG pipeline** combining dense vector search with **BM25** sparse retrieval, fusing results via **Reciprocal Rank Fusion** to improve retrieval relevance over BM25-only search, with citation-grounded answers generated across both **OpenAI** and **Anthropic** models
- Engineered a **FastAPI** backend implementing a four-stage document pipeline (**ingest** → **index** → **retrieve** → **generate**), using **OpenAI embeddings** and **ChromaDB** for vector storage with idempotent chunk upserts to support safe re-ingestion of updated documents
- Designed a parallel retrieval architecture that fires dense and **BM25** sparse searches concurrently, merging results before passing them to a multi-provider **LLM dispatch** layer for answer generation

Codebase Q&A Agent | *Python, Claude API, pgvector, Slack Bolt, VoyageAI*

[GitHub](#)

- Built an **agentic RAG** pipeline where **Claude** autonomously decides when and how to query a vector database via **tool use**, issuing multiple self-refined search queries across up to 4 iterations before producing a grounded answer with file/line citations
- Engineered an ingestion pipeline that clones **GitHub** repositories, chunks source files into overlapping windows, embeds them with **Voyage AI's** code-specialized embedding model, and stores 1024-dim vectors in **PostgreSQL** using **pgvector** for cosine similarity search
- Built **/ask** and **/ingest Slack Bolt** commands enabling real-time codebase Q&A and on-demand repository indexing, containerized with **Docker** for reproducible deployment

PawPath: AI Dog Walking App | *Python, Gemini, Supabase, Google Maps*

[GitHub](#)

- Engineered a **FastAPI** backend with **RESTful endpoints** integrating **Gemini AI** for photo-based dog breed identification and **Google Maps Platform** (Routes, Places, Weather) for dynamic walk route generation
- Built a safety scoring system combining 5 environmental signals (temperature, AQI, UV index, humidity, wind) to assess walk conditions and surface breed-specific health risk alerts
- Implemented full **CRUD** for dog profiles and walk history on **Supabase**, adding live GPS tracking and a breed-specific exercise recommendation engine

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL (PostgreSQL)

AI & ML: LLM Integration, RAG Pipelines, AI Agent Design, Prompt & Context Engineering, Reinforcement Learning, RLHF, Embeddings, Vector Databases, Fine-Tuning (LoRA/QLoRA), Multimodal AI

Frameworks & Libraries: HuggingFace Transformers, LangChain, LangGraph, OpenAI/Anthropic SDKs

Backend & Infrastructure: Node.js, FastAPI, Express, PostgreSQL, Redis, Prisma, Docker, Nginx, JWT, OAuth 2.0, WebSockets

Tools & Platforms: Git, GitHub Actions, Railway, AWS, Vercel, Supabase, Linux